

LOCAL SCALE-INVARIANT CONCENTRATION AND THE TYPE I/TYPE II DICHOTOMY FOR NAVIER-STOKES

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ABSTRACT. We prove a local analytic reduction for possible finite-time singularities of the three-dimensional Navier–Stokes equations. For a suitable weak solution, every singular point at time T has arbitrarily small backward parabolic cylinders on which both the standard mean-subtracted Caffarelli–Kohn–Nirenberg scale-invariant quantity $C + D$ and the scale-invariant velocity quantity C are bounded below by positive constants. If these local scale-invariant quantities tend to zero at each point of the singular set, the solution is regular at time T by the ε -regularity theorem. Along any such positive concentration sequence, the solution either satisfies a local Type I scale-invariant bound or falls under the local Type II alternative.

1. INTRODUCTION

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + u \cdot \nabla u + \nabla p = \Delta u, \quad \nabla \cdot u = 0.$$

We use the standard normalization in which the viscosity is one. The weak solution theory goes back to Leray [4]. Suitable weak solutions and their partial regularity theory were introduced and developed by Scheffer [6] and by Caffarelli, Kohn, and Nirenberg [1]; see also Lin [5]. This paper establishes a local reduction from a possible singularity at time T to the Type I/Type II blow-up rate dichotomy.

Scale-invariant regularity and blow-up criteria provide the standard background for this formulation; see Serrin [8], Giga [3], Escauriaza–Seregin–Šverák [2], and Seregin [7]. The principal regularity input used below is the Caffarelli–Kohn–Nirenberg ε -regularity theorem.

The proof proceeds in four steps. First, the local scale-invariant quantities C and D are finite for suitable weak solutions and invariant under the usual parabolic scaling. Second, the Caffarelli–Kohn–Nirenberg ε -regularity theorem implies that vanishing of $C + D$ at a candidate singular point gives local regularity, while the velocity ε -regularity criterion gives positive concentration in the scale-invariant $L^3_{x,t}$ quantity. Third, a finite-time singularity therefore produces sequences of shrinking cylinders on which $C + D$, and separately C , are bounded below by positive constants. Finally, once such a sequence is fixed, it either satisfies a Type I scale-invariant bound or falls under the local Type II alternative.

All arguments are local in spacetime. Apart from the standard Caffarelli–Kohn–Nirenberg ε -regularity theorem, all quantities and alternatives used in the argument are defined below.

2. SUITABLE WEAK SOLUTIONS AND SINGULAR POINTS

Definition 2.1 (Suitable weak solution). Let $I \subset \mathbb{R}$ be an open interval. A pair (u, p) is a suitable weak solution on $\mathbb{R}^3 \times I$ if

$$u \in L^\infty_t L^2_{x,\text{loc}} \cap L^2_t H^1_{x,\text{loc}}, \quad p \in L^{3/2}_{\text{loc}}, \quad \nabla \cdot u = 0,$$

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the Navier–Stokes equations hold in the sense of distributions, and the local energy inequality holds: for almost every $t_1 < t_2$ in I and every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times I)$,

$$\begin{aligned} & \int_{\mathbb{R}^3} \frac{|u(x, t_2)|^2}{2} \phi(x, t_2) dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \phi dx dt \\ & \leq \int_{\mathbb{R}^3} \frac{|u(x, t_1)|^2}{2} \phi(x, t_1) dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \frac{|u|^2}{2} (\partial_t \phi + \Delta \phi) dx dt \\ & \quad + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p \right) u \cdot \nabla \phi dx dt. \end{aligned}$$

The pressure is understood modulo functions of time.

For $z_0 = (x_0, t_0)$ and $r > 0$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

We also write $Q_r(x_0, t_0)$ for $Q_r((x_0, t_0))$.

Definition 2.2 (Singular set at a fixed time). For a time T , define

$$\Sigma(T) = \{x \in \mathbb{R}^3 : u \text{ is not locally bounded in any } Q_r(x, T)\}.$$

Equivalently, $x \notin \Sigma(T)$ if there are $r > 0$ and $M < \infty$ such that $\|u\|_{L^\infty(Q_r(x, T))} \leq M$.

If a finite-time singularity occurs at time T , then $\Sigma(T) \neq \emptyset$ in this local sense.

3. PARABOLIC RESCALING AND SCALE-INVARIANT QUANTITIES

Fix $z_0 = (x_0, T)$. For $r > 0$, define the parabolic rescaling

$$u^{(r)}(y, s) = r u(x_0 + ry, T + r^2 s), \quad p^{(r)}(y, s) = r^2 p(x_0 + ry, T + r^2 s).$$

The rescaled variables are defined on the set of (y, s) for which $(x_0 + ry, T + r^2 s)$ belongs to the original spacetime domain.

Proposition 3.1 (Invariance under parabolic rescaling). *If (u, p) is a suitable weak solution near (x_0, T) , then $(u^{(r)}, p^{(r)})$ is a suitable weak solution on the rescaled domain. The local energy inequality is preserved, and the pressure remains defined modulo functions of the rescaled time.*

Proof. The distributional equations and the divergence condition follow from the change of variables $x = x_0 + ry$, $t = T + r^2 s$. We include the verification of the local energy inequality to fix the normalization.

Let $\psi \in C_c^\infty$ be nonnegative and compactly supported in the rescaled domain. Define

$$\phi(x, t) = \psi \left(\frac{x - x_0}{r}, \frac{t - T}{r^2} \right).$$

If $y = (x - x_0)/r$ and $s = (t - T)/r^2$, then

$$\partial_t \phi = r^{-2} (\partial_s \psi)(y, s), \quad \nabla_x \phi = r^{-1} (\nabla_y \psi)(y, s), \quad \Delta_x \phi = r^{-2} (\Delta_y \psi)(y, s).$$

For almost every $s_1 < s_2$, the corresponding times $T + r^2 s_1 < T + r^2 s_2$ are admissible in the original local energy inequality. Apply that inequality for (u, p) to ϕ on $[T + r^2 s_1, T + r^2 s_2]$. After the change of variables $x = x_0 + ry$, $t = T + r^2 s$, using

$$u(x_0 + ry, T + r^2 s) = r^{-1} u^{(r)}(y, s), \quad p(x_0 + ry, T + r^2 s) = r^{-2} p^{(r)}(y, s),$$

each term contains the same common factor r . Dividing by this factor gives the rescaled inequality

$$\begin{aligned} & \int_{\mathbb{R}^3} \frac{|u^{(r)}(y, s_2)|^2}{2} \psi(y, s_2) dy + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} |\nabla_y u^{(r)}|^2 \psi dy ds \\ & \leq \int_{\mathbb{R}^3} \frac{|u^{(r)}(y, s_1)|^2}{2} \psi(y, s_1) dy + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \frac{|u^{(r)}|^2}{2} (\partial_s \psi + \Delta_y \psi) dy ds \\ & \quad + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \left(\frac{|u^{(r)}|^2}{2} + p^{(r)} \right) u^{(r)} \cdot \nabla_y \psi dy ds. \end{aligned}$$

Thus $(u^{(r)}, p^{(r)})$ is suitable on the rescaled domain. The pressure transforms according to the Navier–Stokes scaling, and addition of a function of t becomes addition of a function of s . $\square$

Definition 3.2 (Scale-invariant local quantities). For $z_0 = (x_0, t_0)$ and $r > 0$, set

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt$$

and

$$D(z_0, r) = r^{-2} \int_{t_0-r^2}^{t_0} \int_{B_r(x_0)} |p(x, t) - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

Here $(p)_{B_r(x_0)}(t)$ denotes the spatial mean of $p(\cdot, t)$ on $B_r(x_0)$.

The subtraction of the spatial mean removes the pressure ambiguity in the form used by the Caffarelli–Kohn–Nirenberg criterion.

Proposition 3.3 (Local finiteness and scaling). *Let (u, p) be a suitable weak solution near $z_0 = (x_0, T)$. Then $C(z_0, r)$ and $D(z_0, r)$ are finite for every sufficiently small r . Moreover, for every fixed $R < \infty$,*

$$\begin{aligned} & \int_{-R^2}^0 \int_{B_R} \left(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \right) dy ds \\ & = R^2 (C(z_0, rR) + D(z_0, rR)). \end{aligned}$$

Proof. Local finiteness of the velocity term follows from

$$u \in L_t^\infty L_{x,\text{loc}}^2 \cap L_t^2 H_{x,\text{loc}}^1 \subset L_{\text{loc}}^3$$

on finite cylinders. Indeed, if $B \Subset \mathbb{R}^3$ and $(a, b) \Subset I$, Sobolev and interpolation give

$$\int_a^b \|u(t)\|_{L^3(B)}^3 dt \leq C \left(\operatorname{ess\,sup}_{a < t < b} \|u(t)\|_{L^2(B)} \right)^{3/2} \left(\int_a^b \|u(t)\|_{H^1(B)}^2 dt \right)^{3/4} (b-a)^{1/4}.$$

Thus $u \in L_{\text{loc}}^3$. The pressure term is finite because $p \in L_{\text{loc}}^{3/2}$, and subtracting a spatial mean over the same ball preserves local $L^{3/2}$ integrability. More explicitly, for a ball B ,

$$|p - (p)_B(t)|^{3/2} \leq C \left(|p|^{3/2} + |(p)_B(t)|^{3/2} \right),$$

while Jensen's inequality gives

$$|(p)_B(t)|^{3/2} \leq \frac{1}{|B|} \int_B |p(x, t)|^{3/2} dx.$$

After integration over B and over time, the mean-subtracted pressure term is controlled by the local $L^{3/2}$ norm of p .

For the scaling identity, use $x = x_0 + ry$, $t = T + r^2s$. The velocity term transforms as

$$\int_{-R^2}^0 \int_{B_R} |u^{(r)}|^3 dy ds = r^{-2} \int_{T-(rR)^2}^T \int_{B_{rR}(x_0)} |u|^3 dx dt.$$

The pressure means transform according to

$$(p^{(r)})_{B_R}(s) = r^2(p)_{B_{rR}(x_0)}(T + r^2s),$$

because

$$\frac{1}{|B_R|} \int_{B_R} r^2 p(x_0 + ry, T + r^2s) dy = \frac{r^2}{|B_{rR}|} \int_{B_{rR}(x_0)} p(x, T + r^2s) dx.$$

Therefore

$$\begin{aligned} & \int_{-R^2}^0 \int_{B_R} |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} dy ds \\ &= r^{-2} \int_{T-(rR)^2}^T \int_{B_{rR}(x_0)} |p - (p)_{B_{rR}(x_0)}(t)|^{3/2} dx dt. \end{aligned}$$

Adding the velocity and pressure identities gives the displayed formula. $\square$

4. ε -REGULARITY AND THE VANISHING CASE

Theorem 4.1 (Caffarelli–Kohn–Nirenberg ε -regularity). *There exists a universal constant $\varepsilon_0 > 0$ with the following property. Let (u, p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) + D(z_0, r) < \varepsilon_0,$$

then u is locally bounded in a smaller cylinder, for instance in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. This is the local ε -regularity theorem of Caffarelli–Kohn–Nirenberg [1]; see also Lin [5]. It is stated here with the spatial mean subtraction used in D . Equivalent representatives of the pressure differ by functions of time, and the spatial mean subtraction on $B_r(x_0)$ removes this ambiguity. $\square$

Theorem 4.2 (Velocity ε -regularity). *There exists a universal constant $\varepsilon_v > 0$ with the following property. Let (u, p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) < \varepsilon_v,$$

then u is locally bounded in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. This is the standard $L^3_{x,t}$ form of the Caffarelli–Kohn–Nirenberg ε -regularity criterion. It follows from the standard local pressure decay lemma together with Theorem 4.1; equivalently, it is the $L^3_{x,t}$ -smallness criterion proved in the Caffarelli–Kohn–Nirenberg theory and in Lin’s direct proof [1, 5]. The statement is scale invariant because C is invariant under the parabolic Navier–Stokes scaling. $\square$

Definition 4.3 (Vanishing local scale-invariant quantities at time T). We say that the local scale-invariant quantities vanish at time T if, for every $x_0 \in \Sigma(T)$,

$$\limsup_{r \downarrow 0} (C((x_0, T), r) + D((x_0, T), r)) = 0.$$

This condition is vacuous when $\Sigma(T) = \emptyset$.

Proposition 4.4 (Equivalent rescaled formulation). *The preceding condition is equivalent to the following statement: for every $x_0 \in \Sigma(T)$ and every fixed $R < \infty$,*

$$\lim_{r \downarrow 0} \int_{-R^2}^0 \int_{B_R} \left(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \right) dy ds = 0.$$

Proof. By Proposition 3.3, the rescaled integral over $B_R \times (-R^2, 0)$ equals

$$R^2 (C((x_0, T), rR) + D((x_0, T), rR)).$$

Thus vanishing of $C + D$ implies vanishing on every fixed rescaled cylinder. Conversely, taking $R = 1$ gives the original pointwise condition. $\square$

Theorem 4.5 (Regularity under vanishing local quantities). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ for some $\delta > 0$. If the local scale-invariant quantities vanish at time T , then $\Sigma(T) = \emptyset$. Thus no finite-time singularity occurs at time T in the no-concentration case.*

Proof. Suppose, toward a contradiction, that $x_0 \in \Sigma(T)$. By the vanishing hypothesis, choose $r > 0$ such that

$$C((x_0, T), r) + D((x_0, T), r) < \varepsilon_0.$$

Theorem 4.1 implies that u is bounded in $Q_{r/2}(x_0, T)$, hence $x_0 \notin \Sigma(T)$. This contradicts the choice of x_0 . Hence $\Sigma(T) = \emptyset$. $\square$

Corollary 4.6 (Vanishing local quantities exclude a terminal singularity). *Let (u, p) be suitable near (x_0, T) . If*

$$\limsup_{r \downarrow 0} (C((x_0, T), r) + D((x_0, T), r)) = 0,$$

then u is bounded in some backward cylinder $Q_\rho(x_0, T)$, and hence $x_0 \notin \Sigma(T)$. Equivalently, membership in $\Sigma(T)$ requires that $C + D$ have a positive lower bound along a sequence of shrinking backward parabolic cylinders.

Proof. The hypothesis gives a scale at which $C + D$ is below the universal threshold in Theorem 4.1. The ε -regularity theorem gives local boundedness in a backward cylinder ending at T , so $x_0 \notin \Sigma(T)$. $\square$

5. POSITIVE LOCAL CONCENTRATION

Lemma 5.1 (Failure of vanishing gives a positive concentration sequence). *Suppose the local scale-invariant quantities do not vanish at time T . Then there exist $x_0 \in \Sigma(T)$, a constant $\eta > 0$, and radii $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Equivalently, the rescaled sequence around (x_0, T) has a positive lower bound for the corresponding local scale-invariant quantities on a fixed bounded cylinder.

Proof. Negating the definition of vanishing gives a point $x_0 \in \Sigma(T)$ with positive limsup of

$$C((x_0, T), r) + D((x_0, T), r)$$

as $r \downarrow 0$. Choose $\eta > 0$ below that limsup and choose a sequence $r_n \downarrow 0$ along which the displayed lower bound holds. The rescaled formulation follows from Proposition 3.3. $\square$

Definition 5.2 (Positive local concentration sequence). Let $x_0 \in \Sigma(T)$. A sequence $r_n \downarrow 0$ is a positive local concentration sequence at (x_0, T) if there is $\eta > 0$ such that

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

The corresponding parabolic rescalings are those defined by

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Proposition 5.3 (Persistence of positive concentration). *Let (u_n, p_n) be suitable weak solutions on a fixed cylinder $B_R \times (-R^2, 0)$. Suppose that*

$$\int_{-R^2}^0 \int_{B_R} \left(|u_n|^3 + |p_n - (p_n)_{B_R}(s)|^{3/2} \right) dy ds \geq \eta > 0$$

for all n . Suppose further that, after passing to a subsequence,

$$u_n \rightarrow U \quad \text{in } L^3(B_R \times (-R^2, 0))$$

and

$$p_n - (p_n)_{B_R}(s) \rightarrow \Pi \quad \text{in } L^{3/2}(B_R \times (-R^2, 0)).$$

Then

$$\int_{-R^2}^0 \int_{B_R} \left(|U|^3 + |\Pi|^{3/2} \right) dy ds \geq \eta.$$

In particular, the limiting pair retains the same positive lower bound for the mean-subtracted local scale-invariant quantity. If $\Pi = P - (P)_{B_R}(s)$, this is the corresponding Caffarelli–Kohn–Nirenberg quantity of (U, P) on $B_R \times (-R^2, 0)$.

Proof. Strong convergence in L^3 gives

$$\int_{-R^2}^0 \int_{B_R} |u_n|^3 dy ds \longrightarrow \int_{-R^2}^0 \int_{B_R} |U|^3 dy ds.$$

Strong convergence in $L^{3/2}$ gives the analogous convergence of the mean-subtracted pressure integrals. Passing to the limit in the lower bound yields the asserted inequality. $\square$

Remark 5.4. [Proposition 5.3](#) is an auxiliary compactness observation, independent of the proof of the dichotomy below.

Proposition 5.5 (Local alternative). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ for some $\delta > 0$. Exactly one of the following alternatives holds.*

(i) For every $x_0 \in \Sigma(T)$,

$$\limsup_{r \downarrow 0} (C((x_0, T), r) + D((x_0, T), r)) = 0.$$

(ii) There exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Proof. The first alternative is a universal assertion over the singular set, and the second is its failure expressed by a sequence. If the first alternative fails, then for some $x_0 \in \Sigma(T)$ the corresponding limsup is positive; choosing a positive number below this limsup and a sequence along which the lower bound holds gives the second alternative. Conversely, the second alternative contradicts the first. Hence exactly one alternative holds. $\square$

Theorem 5.6 (Finite-time singularity yields positive local scale-invariant concentration). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. If $\Sigma(T) \neq \emptyset$, then there are $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Proof. If the first alternative in [Proposition 5.5](#) held, then [Theorem 4.5](#) would imply $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore the second alternative in [Proposition 5.5](#) holds. $\square$

Corollary 5.7 (Singular points carry positive velocity concentration). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. If $x_0 \in \Sigma(T)$, then there are $\eta_v > 0$ and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) \geq \eta_v \quad \text{for all } n.$$

Proof. If the conclusion failed, then

$$\limsup_{r \downarrow 0} C((x_0, T), r) = 0.$$

Choose $r > 0$ so small that

$$C((x_0, T), r) < \varepsilon_v,$$

where ε_v is the constant in [Theorem 4.2](#). The velocity ε -regularity theorem gives boundedness of u in $Q_{r/2}(x_0, T)$, contradicting $x_0 \in \Sigma(T)$. Hence the limsup is positive; taking any η_v below it and passing to a sequence with $C((x_0, T), r_n) \geq \eta_v$ gives the result. $\square$

6. THE TYPE I/TYPE II BLOW-UP RATE DICHOTOMY

Let $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ be given by [Theorem 5.6](#). Define

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Then (u_n, p_n) are suitable weak solutions on expanding backward cylinders, and by [Proposition 3.3](#) they retain a positive lower bound for the mean-subtracted local scale-invariant quantity on a fixed bounded cylinder.

Definition 6.1 (Type I concentration sequence). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0, T) . The sequence is called Type I if the concentration point (x_0, T) satisfies a local Type I scale-invariant bound: there are $\rho > 0$ and $M < \infty$, with $\rho^2 < \delta$, such that

$$\operatorname{ess\,sup}_{T - \rho^2 < t < T} \sqrt{T - t} \|u(t)\|_{L^\infty(B_\rho(x_0))} \leq M.$$

This is the Type I alternative.

Definition 6.2 (Local Type II alternative). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0, T) . The sequence belongs to the local Type II alternative if the concentration point (x_0, T) satisfies no local Type I scale-invariant bound. Equivalently, for every $\rho > 0$ with $\rho^2 < \delta$,

$$\operatorname{ess\,sup}_{T - \rho^2 < t < T} \sqrt{T - t} \|u(t)\|_{L^\infty(B_\rho(x_0))} = \infty.$$

Theorem 6.3 (Local blow-up rate dichotomy). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then there exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

The sequence r_n is a positive local concentration sequence. The corresponding concentration regime satisfies either the local Type I bound or the local Type II alternative. Consequently, once the vanishing case has been excluded by ε -regularity, every finite-time singular case falls under exactly one of the two blow-up rate alternatives.

Proof. If the local scale-invariant quantities vanished at every point of $\Sigma(T)$, then [Theorem 4.5](#) would give $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore [Theorem 5.6](#) gives $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ with the stated lower bound. The rescaled sequence has a positive local scale-invariant lower bound by [Proposition 3.3](#). By definition, either the Type I bound in [Definition 6.1](#) holds, or the local Type II alternative in [Definition 6.2](#) holds. $\square$

Remark 6.4. This theorem is local. Its conclusion is the existence of a positive mean-subtracted scale-invariant concentration sequence and the dichotomy between a local Type I bound and failure of every such bound at the concentration point.

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